**Assignment: XML Schema Generation and Validation (2026-10-20)**

**Overview**

In this assignment, you will explore the relationship between XML documents and XML Schemas (XSD). You are required to design an XML document, implement an automated generator to produce an XSD file based on that XML, and build a validator to verify the XML against the generated schema.

**Objectives & Requirements**

1. **Design an XML File (data.xml)**
    - Create a well-formed XML document representing a realistic dataset or scenario (e.g., library catalog, course management system, e-commerce order).
    - The document should include nested child elements and attributes to demonstrate various data structures and data types (e.g., text, integers, booleans, decimals).
2. **Develop an XML Schema Generator (SchemaGenerator)**
    - Write a program (in Java, Python, C++, or another approved language) that reads data.xml as input.
    - Analyze the XML structure, element hierarchy, and attribute types.
    - Automatically generate a corresponding XML Schema file and save it as schema.xsd.
    - The generated schema should include element definitions, attribute declarations, appropriate basic types (e.g., xs:string, xs:integer), and sequence/complex type structures.
3. **Develop an XML Schema Validator (SchemaValidator)**
    - Write a validator program that takes both data.xml and schema.xsd as inputs.

- Validate whether data.xml strictly conforms to schema.xsd.
- Display clear output messages:
  - Indicate **successful validation** if the XML conforms to the schema.
  - Report **validation errors** (with line numbers or error descriptions) if the XML contains schema violations.

**Deliverables**

Please compress all deliverables into a single zip archive named [YourStudentID]_Assignment.zip containing:

1. **Source Code:**
   - XML Schema Generator program source code.
   - XML Schema Validator program source code.
2. **Data & Schema Files:**
   - data.xml (original XML dataset).
   - schema.xsd (the generated schema file).
   - *(Optional)* A modified/invalid XML test file used to demonstrate failure handling.
3. **Execution Report / README:**
   - Brief compilation and execution instructions.
   - Screenshots demonstrating:
     - The XSD generation process.
     - Successful validation of the valid XML file.
     - Failed validation when an intentional error is introduced.